

Bode Plot

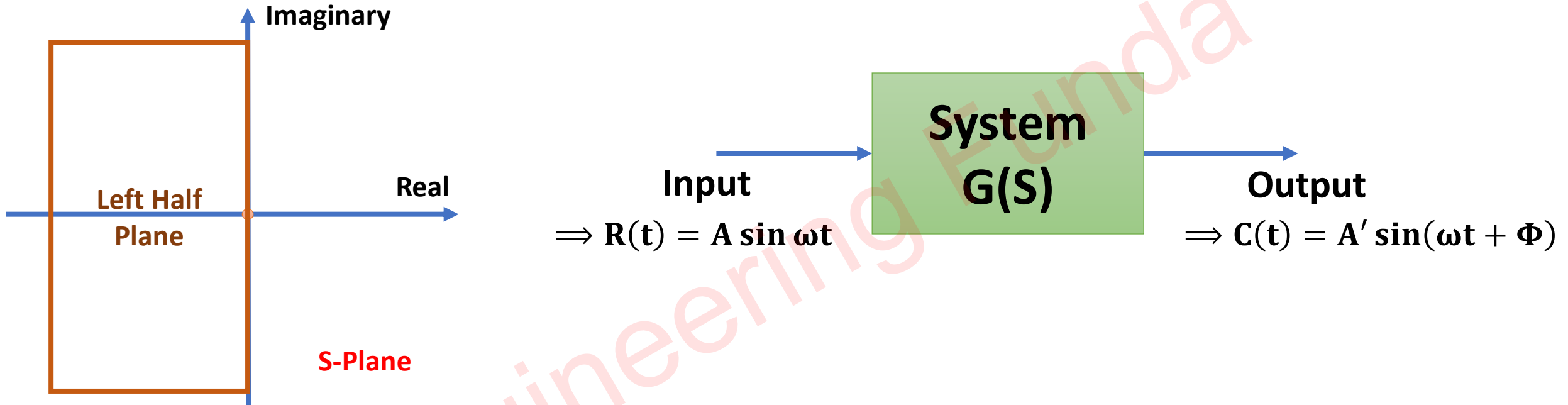
Outlines

- ❑ Basics of Bode Plot
- ❑ Procedure of Bode Plot
- ❑ Parameters of Bode Plot
 - Gain Margin
 - Phase Margin
 - Gain Cross Over Frequency
 - Phase Cross Over Frequency
- ❑ Stability by Bode Plot
- ❑ Advantages of Bode Plot

Basics of Bode Plot

❑ Bode Plot is Only applicable to Minimum Phase Transfer Function

- MPTF – All Roots {Poles/Zeros} are there in Left Half of S Plane.



❑ In Bode Plot, we have two different plots:

- Gain Plot [$|G(j\omega)| \rightarrow \omega$]
- Phase Plot [$\angle G(j\omega) \rightarrow \omega$]

Procedure of Bode Plot

□ Step-1: Write the given Transfer Function in Standard Form.

- Example: If the System $G(S)$ is given by: $\Rightarrow G(S) = \frac{(S+a)(S+b)}{(S+p)(S+q)}$
- $G(S)$ in Standard Form is given by: $\Rightarrow G(S) = \frac{ab}{pq} \frac{\left(1+\frac{S}{a}\right)\left(1+\frac{S}{b}\right)}{\left(1+\frac{S}{p}\right)\left(1+\frac{S}{q}\right)}$, Where, $k = \frac{ab}{pq}$

□ Step-2: Identify Slope of 1st Line. It is based on Poles/Zeros at Origin.

- For no Poles/Zeros at Origin, Slope of 1st is 0 dB/Dec.
- For One Pole at Origin, Slope of 1st = -20 dB/Dec.
- For Two Poles at Origin, Slope of 1st = -40 dB/Dec, and so on.
- For One Zero at Origin, Slope of 1st = +20 dB/Dec.
- For Two Zeros at Origin, Slope of 1st = +40 dB/Dec, and so on.

□ Step-3: Identify Gain of 1st Line at $\omega = 1$ rad/sec.

- $\text{Gain}|_{\omega=1 \text{ rad/sec}} = 20 \log k$

□ **Step-4: Write all corner frequencies in ascending Order and define the slope for each line.**

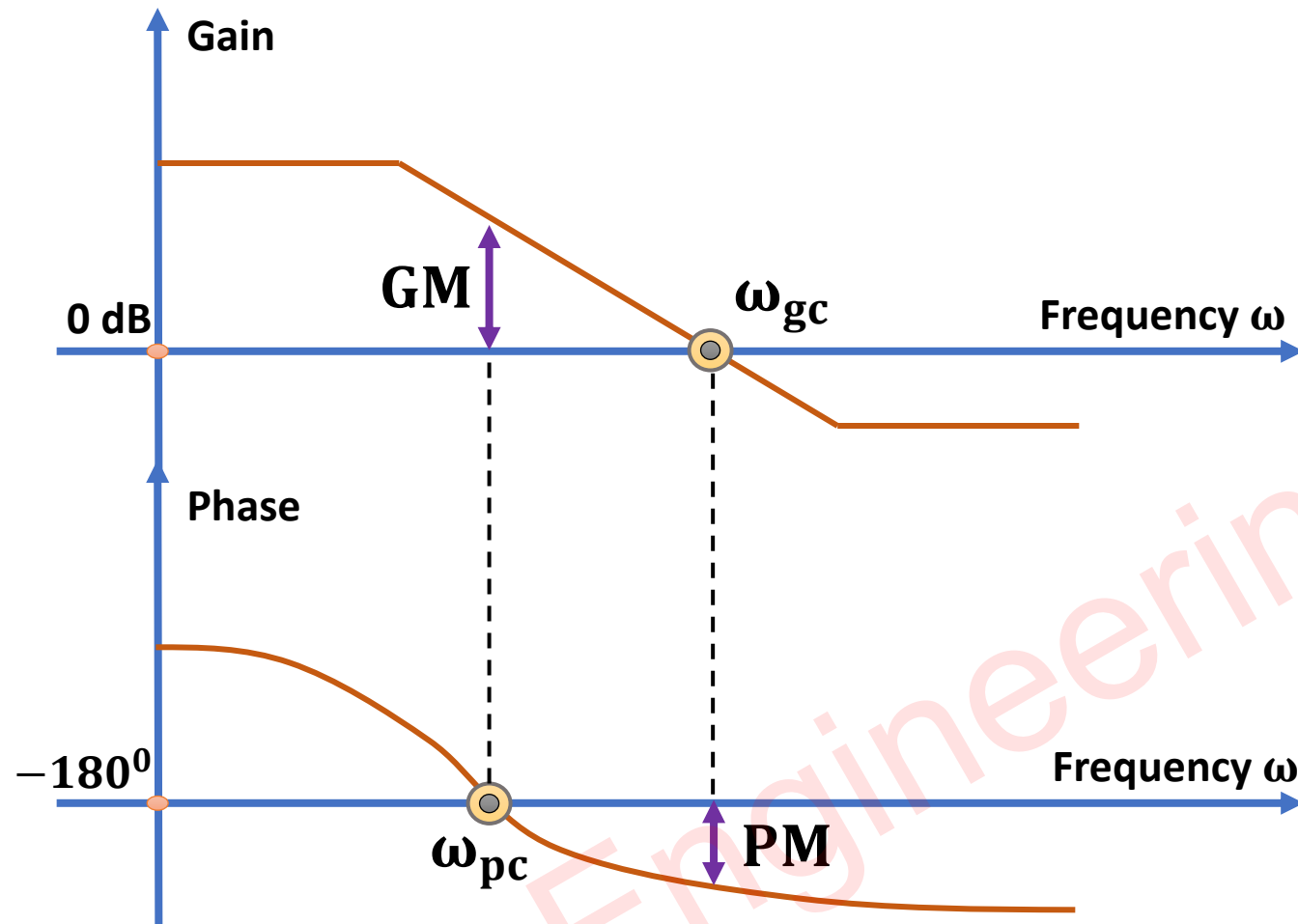
- **Example: If the System $G(S)$ is given by: $\Rightarrow G(S) = \frac{(S+a)(S+b)}{(S+p)(S+q)}$**
- **If $a < p < b < q$**
- **Slope of 1st Line = 0 dB/Dec**

Corner Frequency	Pole/Zero	Slope	Resultant Slope
a	Zero	+20 dB/Dec	= 20 dB/Dec
p	Pole	-20 dB/Dec	= 0 dB/Dec
b	Zero	+20 dB/Dec	= 20 dB/Dec
q	Pole	-20 dB/Dec	= 0 dB/Dec

□ **Step-5: Phase Plot of $\angle G(j\omega)$.**

- **Example: If the System $G(j\omega)$ is given by: $\Rightarrow G(j\omega) = \frac{(j\omega+a)(j\omega+b)}{(j\omega+p)(j\omega+q)}$**
- **$\angle G(j\omega) = \tan^{-1}\left(\frac{\omega}{a}\right) + \tan^{-1}\left(\frac{\omega}{b}\right) - \tan^{-1}\left(\frac{\omega}{p}\right) - \tan^{-1}\left(\frac{\omega}{q}\right)$**

Parameters of Bode Plot



- ❑ **Gain Cross Over Frequency:** It is a frequency at which Gain Crosses 0 dB.
- ❑ **Phase Cross Over Frequency:** It is a frequency at which Phase Crosses -180° .
- ❑ **Gain Margin:** It is a margin of Gain with respect to 0 dB at Phase Cross Over Frequency.
 - Above 0 dB, Gain Margin is -Ve and Below 0 dB, Gain Margin is +Ve.
- ❑ **Phase Margin:** It is a margin of Phase with respect to -180° at Gain Cross Over Frequency.
 - Above -180° , Phase Margin is +Ve and Below -180° , Phase Margin is -Ve.

Stability using Bode Plot

- ❑ For Stable System: $\omega_{pc} > \omega_{gc}$
- ❑ For Stable System Gain Margin and Phase Margin is +Ve.

Advantages of Bode Plot

- ❑ Frequency Response Representation is Easy by Separate Magnitude and Phase.
- ❑ Logarithmic Scale Covers Broad Range of Frequency.
- ❑ Phase and Gain Information {Phase and Gain Margin, Phase and Gain Cross Frequency}.
- ❑ Stability Analysis.

Engineering Funda

Bode Plot Example

Example 1 — Draw Bode Plot for the Transfer Function:

$$\Rightarrow G(s)H(s) = \frac{100(s + 3)}{s(s + 1)(s + 5)}$$

Also Find,

1. Gain Cross Over Frequency
2. Phase Cross Over Frequency
3. Gain Margin
4. Phase Margin
5. Stability

□ Step-1: Write the given Transfer Function in Standard Form.

$$\Rightarrow G(s)H(s) = \frac{100 \times 3(1 + s/3)}{5 \times s(1 + s)(1 + s/5)} = \frac{60(1 + s/3)}{s(1 + s)(1 + s/5)}$$

□ Step-2: Slope of 1st Line. Here, we have one pole at the Origin, Slope of 1st Line = -20 dB/Dec

□ Step-3: Gain of 1st Line at 1 rad/sec.

$$\Rightarrow \text{Gain} \Big|_{\omega=1 \text{ rad/sec}} = 20 \log k = 20 \log 60 = 35.5 \text{ dB}$$

❑ Step-4: Write all corner frequency in ascending Order and define the slope for each line.

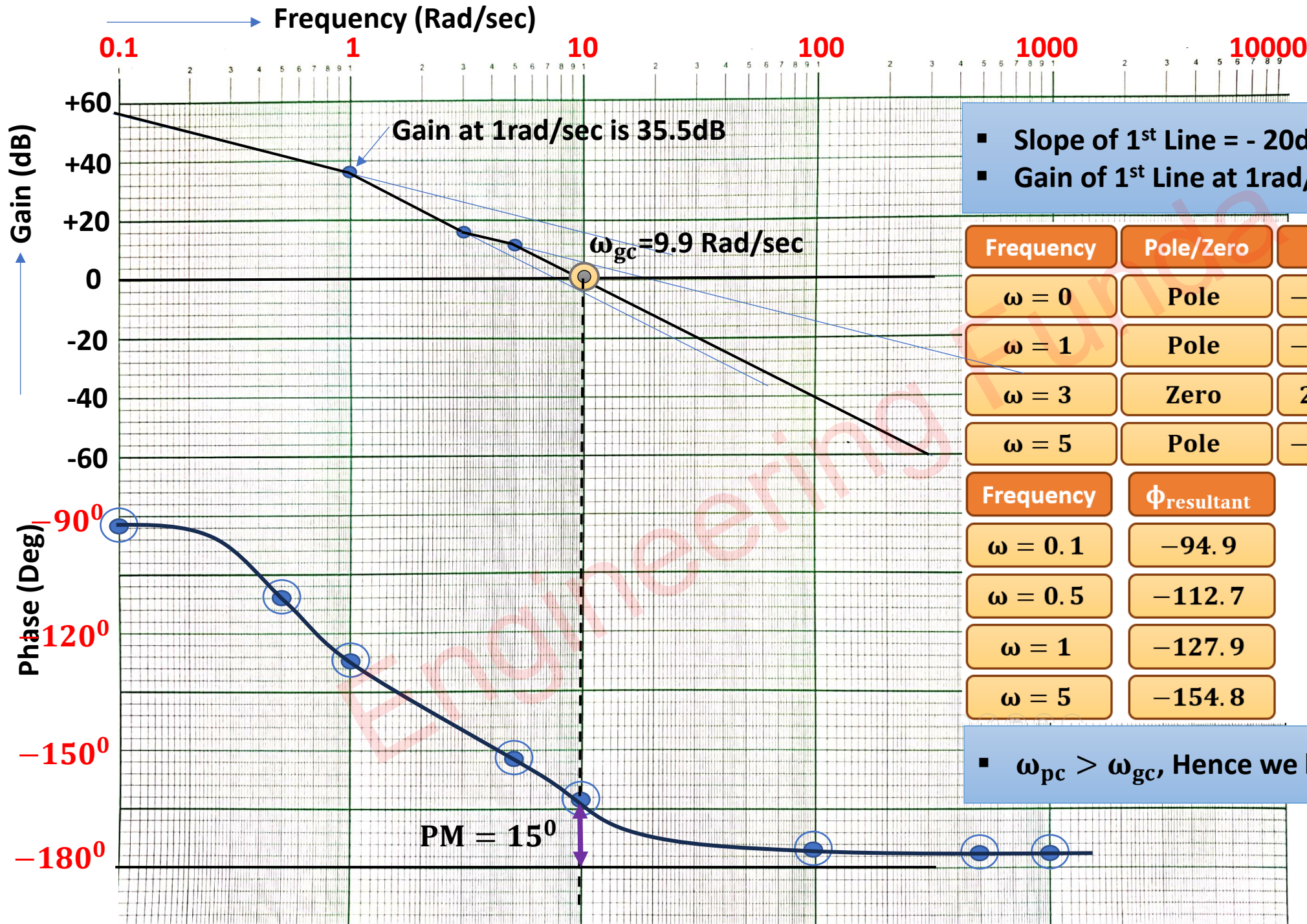
$$\Rightarrow G(s)H(s) = \frac{100(s + 3)}{s(s + 1)(s + 5)}$$

Frequency	Pole/Zero	Slope	Resultant
$\omega = 0$	Pole	-20dB/Dec	-20dB/Dec
$\omega = 1$	Pole	-20dB/Dec	-40dB/Dec
$\omega = 3$	Zero	20dB/Dec	-20dB/Dec
$\omega = 5$	Pole	-20dB/Dec	-40dB/Dec

❑ Step-5: Phase of $\angle G(j\omega)$.

$$\Rightarrow \angle G(j\omega) = \phi_{\text{resultant}} = -90 - \tan^{-1} \omega - \tan^{-1} \frac{\omega}{5} + \tan^{-1} \frac{\omega}{3}$$

Frequency	$\phi_{\text{resultant}}$	Frequency	$\phi_{\text{resultant}}$
$\omega = 0.1$	-94.9	$\omega = 10$	-164.4
$\omega = 0.5$	-112.7	$\omega = 100$	-178.2
$\omega = 1$	-127.9	$\omega = 500$	-179.6
$\omega = 5$	-154.8	$\omega = 1000$	-179.8



- Slope of 1st Line = - 20dB/Dec.
- Gain of 1st Line at 1rad/sec = $20 \log k = 35.5\text{dB}$

Frequency	Pole/Zero	Slope	Resultant
$\omega = 0$	Pole	-20dB/Dec	-20dB/Dec
$\omega = 1$	Pole	-20dB/Dec	-40dB/Dec
$\omega = 3$	Zero	20dB/Dec	-20dB/Dec
$\omega = 5$	Pole	-20dB/Dec	-40dB/Dec

Frequency	$\phi_{\text{resultant}}$	Frequency	$\phi_{\text{resultant}}$
$\omega = 0.1$	-94.9	$\omega = 10$	-164.4
$\omega = 0.5$	-112.7	$\omega = 100$	-178.2
$\omega = 1$	-127.9	$\omega = 500$	-179.6
$\omega = 5$	-154.8	$\omega = 1000$	-179.8

- $\omega_{pc} > \omega_{gc}$, Hence we have a Stable System.

Bode Plot Example

Example 2 — Draw Bode Plot for the Transfer Function:

$$\Rightarrow G(s) = \frac{14400(s + 5)}{s^2(s + 20)(s + 100)}$$

Also Find,

1. Gain Cross Over Frequency
2. Phase Cross Over Frequency
3. Gain Margin
4. Phase Margin
5. Stability

□ Step-1: Write the given Transfer Function in Standard Form.

$$\Rightarrow G(s) = \frac{14400 \times 5(1 + s/5)}{20 \times 100 \times s^2(1 + s/20)(1 + s/100)} = \frac{36(1 + s/5)}{s^2(1 + s/20)(1 + s/100)}$$

□ Step-2: Slope of 1st Line. Here, we have two poles at the Origin, Slope of 1st Line = -40 dB/Dec

□ Step-3: Gain of 1st Line at 1 rad/sec.

$$\Rightarrow \text{Gain} \Big|_{\omega=1 \text{ rad/sec}} = 20 \log k = 20 \log 36 = 31.1 \text{ dB}$$

❑ Step-4: Write all corner frequency in ascending Order and define the slope for each line.

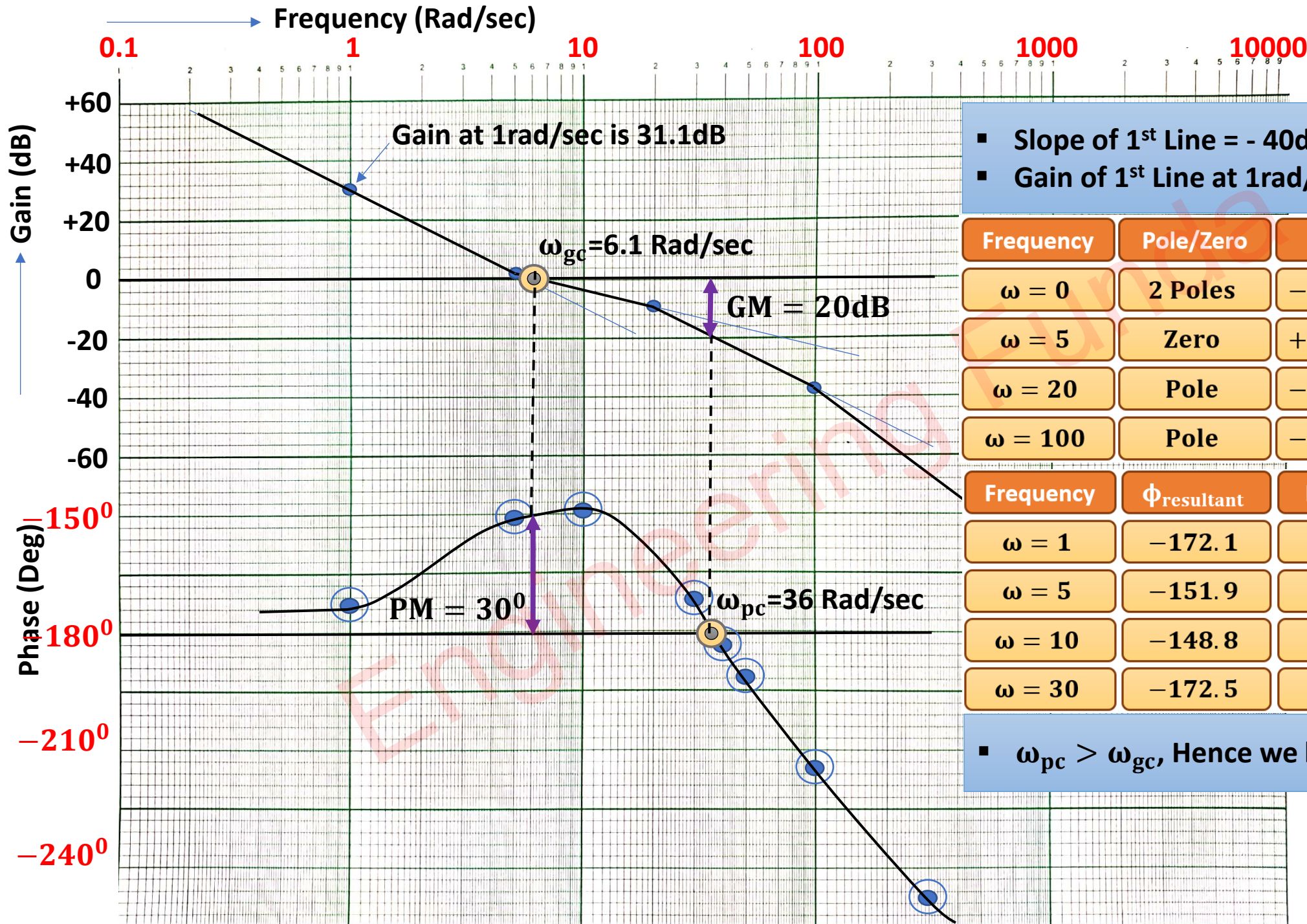
$$\Rightarrow G(s) = \frac{14400(s + 5)}{s^2(s + 20)(s + 100)}$$

Frequency	Pole/Zero	Slope	Resultant
$\omega = 0$	2 Poles	-40dB/Dec	-40dB/Dec
$\omega = 5$	Zero	$+20\text{dB/Dec}$	-20dB/Dec
$\omega = 20$	Pole	-20dB/Dec	-40dB/Dec
$\omega = 100$	Pole	-20dB/Dec	-60dB/Dec

❑ Step-5: Phase of $\angle G(j\omega)$.

$$\Rightarrow \angle G(j\omega) = \phi_{\text{resultant}} = -180 - \tan^{-1} \frac{\omega}{20} - \tan^{-1} \frac{\omega}{100} + \tan^{-1} \frac{\omega}{5}$$

Frequency	$\phi_{\text{resultant}}$	Frequency	$\phi_{\text{resultant}}$
$\omega = 1$	-172.1	$\omega = 40$	-182.4
$\omega = 5$	-151.9	$\omega = 50$	-190.5
$\omega = 10$	-148.8	$\omega = 100$	-216.6
$\omega = 30$	-172.5	$\omega = 300$	-248.7



- Slope of 1st Line = - 40dB/Dec.
- Gain of 1st Line at 1rad/sec = $20 \log k = 31.1\text{dB}$

Frequency	Pole/Zero	Slope	Resultant
$\omega = 0$	2 Poles	-40dB/Dec	-40dB/Dec
$\omega = 5$	Zero	+20dB/Dec	-20dB/Dec
$\omega = 20$	Pole	-20dB/Dec	-40dB/Dec
$\omega = 100$	Pole	-20dB/Dec	-60dB/Dec

Frequency	$\phi_{\text{resultant}}$	Frequency	$\phi_{\text{resultant}}$
$\omega = 1$	-172.1	$\omega = 40$	-182.4
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$\omega = 30$	-172.5	$\omega = 300$	-248.7

- $\omega_{pc} > \omega_{gc}$, Hence we have a Stable System.

Bode Plot Example

Example 3 — Draw Bode Plot for the Transfer Function:

$$\Rightarrow G(s) = \frac{10e^{-0.1s}}{s(s+1)(s+10)}$$

Also Find,

1. Gain Cross Over Frequency
2. Phase Cross Over Frequency
3. Gain Margin
4. Phase Margin
5. Stability

□ Step-1: Write the given Transfer Function in Standard Form.

$$\Rightarrow G(s) = \frac{10e^{-0.1s}}{10 \times s(1+s)(1+s/10)} = \frac{e^{-0.1s}}{s(1+s)(1+s/10)}$$

□ Step-2: Slope of 1st Line. Here, we have one pole at the Origin, Slope of 1st Line = -20 dB/Dec

□ Step-3: Gain of 1st Line at 1 rad/sec.

$$\Rightarrow \text{Gain} \Big|_{\omega=1 \text{ rad/sec}} = 20 \log k = 20 \log 1 = 0 \text{ dB}$$

❑ Step-4: Write all corner frequency in ascending Order and define the slope for each line.

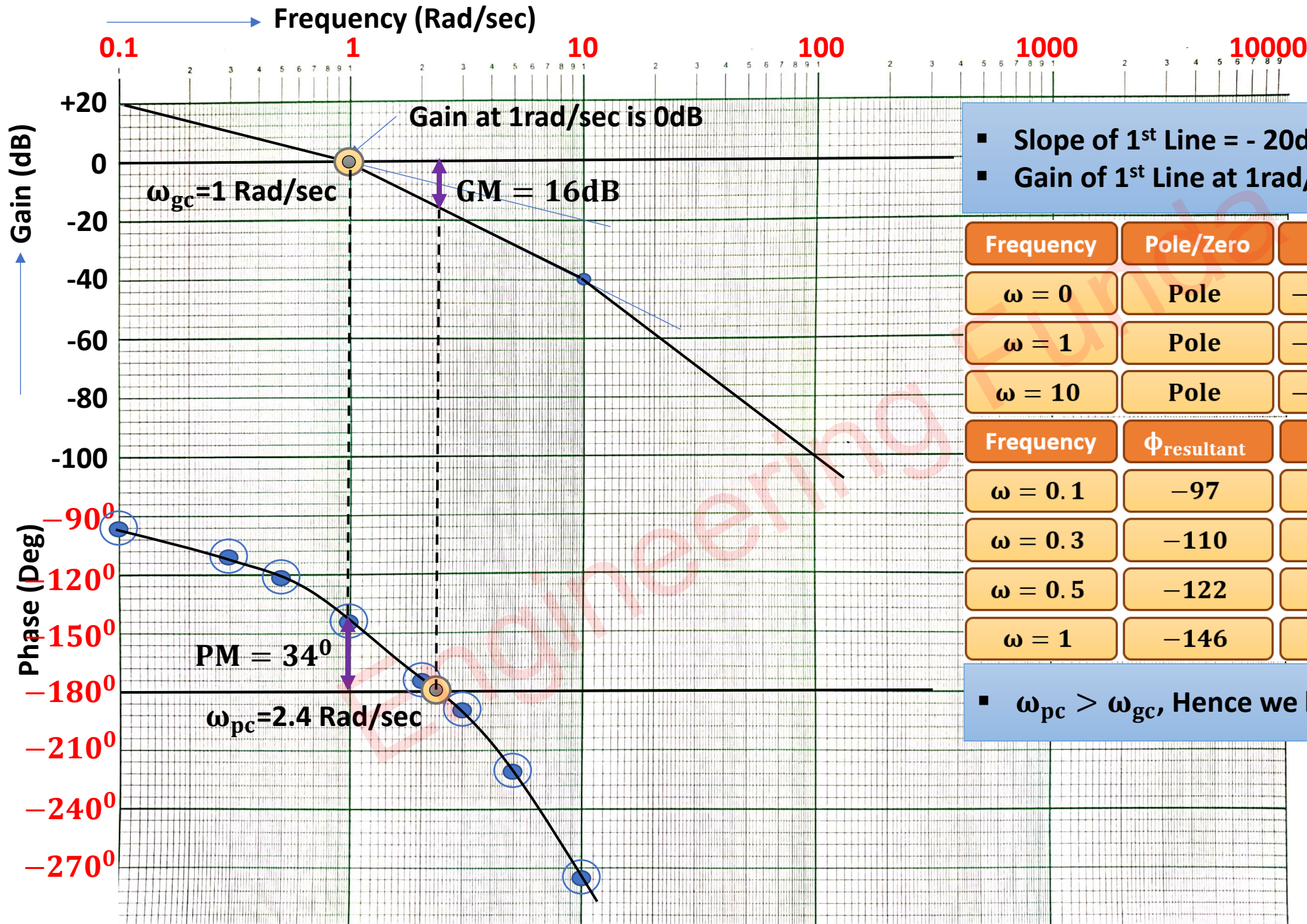
$$\Rightarrow G(s) = \frac{10e^{-0.1s}}{s(s+1)(s+10)}$$

Frequency	Pole/Zero	Slope	Resultant
$\omega = 0$	Pole	-20dB/Dec	-20dB/Dec
$\omega = 1$	Pole	-20dB/Dec	-40dB/Dec
$\omega = 10$	Pole	-20dB/Dec	-60dB/Dec

❑ Step-5: Phase of $\angle G(j\omega)$.

$$\Rightarrow \angle G(j\omega) = \phi_{\text{resultant}} = -90^\circ - \tan^{-1} \omega - \tan^{-1} \frac{\omega}{10} - 0.1\omega$$

Frequency	$\phi_{\text{resultant}}$	Frequency	$\phi_{\text{resultant}}$
$\omega = 0.1$	-97	$\omega = 2$	-176
$\omega = 0.3$	-110	$\omega = 3$	-190
$\omega = 0.5$	-122	$\omega = 5$	-224
$\omega = 1$	-146	$\omega = 10$	-276



- Slope of 1st Line = - 20dB/Dec.
- Gain of 1st Line at 1rad/sec = $20 \log k = 0\text{dB}$

Frequency	Pole/Zero	Slope	Resultant
$\omega = 0$	Pole	-20dB/Dec	-20dB/Dec
$\omega = 1$	Pole	-20dB/Dec	-40dB/Dec
$\omega = 10$	Pole	-20dB/Dec	-60dB/Dec
Frequency	$\phi_{\text{resultant}}$	Frequency	$\phi_{\text{resultant}}$
$\omega = 0.1$	-97	$\omega = 2$	-176
$\omega = 0.3$	-110	$\omega = 3$	-190
$\omega = 0.5$	-122	$\omega = 5$	-224
$\omega = 1$	-146	$\omega = 10$	-276

- $\omega_{pc} > \omega_{gc}$, Hence we have a Stable System.

Bode Plot Example

Example 4 — Draw Bode Plot for the Transfer Function:

$$\Rightarrow G(s) = \frac{64(s + 2)}{s(s + 0.5)(s^2 + 3.2s + 64)}$$

Also Find,

1. Gain Cross Over Frequency
2. Phase Cross Over Frequency
3. Gain Margin
4. Phase Margin
5. Stability

□ **Step-1:** Write the given Transfer Function in Standard Form.

$$\Rightarrow G(s) = \frac{64 \times 2 \left(1 + \frac{s}{2}\right)}{64 \times 0.5 \times s \left(1 + \frac{s}{0.5}\right) \left(1 + \frac{3.2s}{64} + \frac{s^2}{64}\right)} = \frac{4 \left(1 + \frac{s}{2}\right)}{s(1 + 2s) \left(1 + 0.05s + \frac{s^2}{64}\right)}$$

□ **Step-2:** Slope of 1st Line. Here, we have one pole at the Origin, Slope of 1st Line = -20 dB/Dec

□ **Step-3:** Gain of 1st Line at 1 rad/sec.

$$\Rightarrow \text{Gain} \Big|_{\omega=1 \text{ rad/sec}} = 20 \log k = 20 \log 4 = 12\text{dB}$$

❑ Step-4: Write all corner frequency in ascending Order and define the slope for each line.

$$\Rightarrow G(s) = \frac{64(s + 2)}{s(s + 0.5)(s^2 + 3.2s + 64)} \Rightarrow G(j\omega) = \frac{64(j\omega + 2)}{j\omega(j\omega + 0.5)(64 - \omega^2 + j3.2\omega)}$$

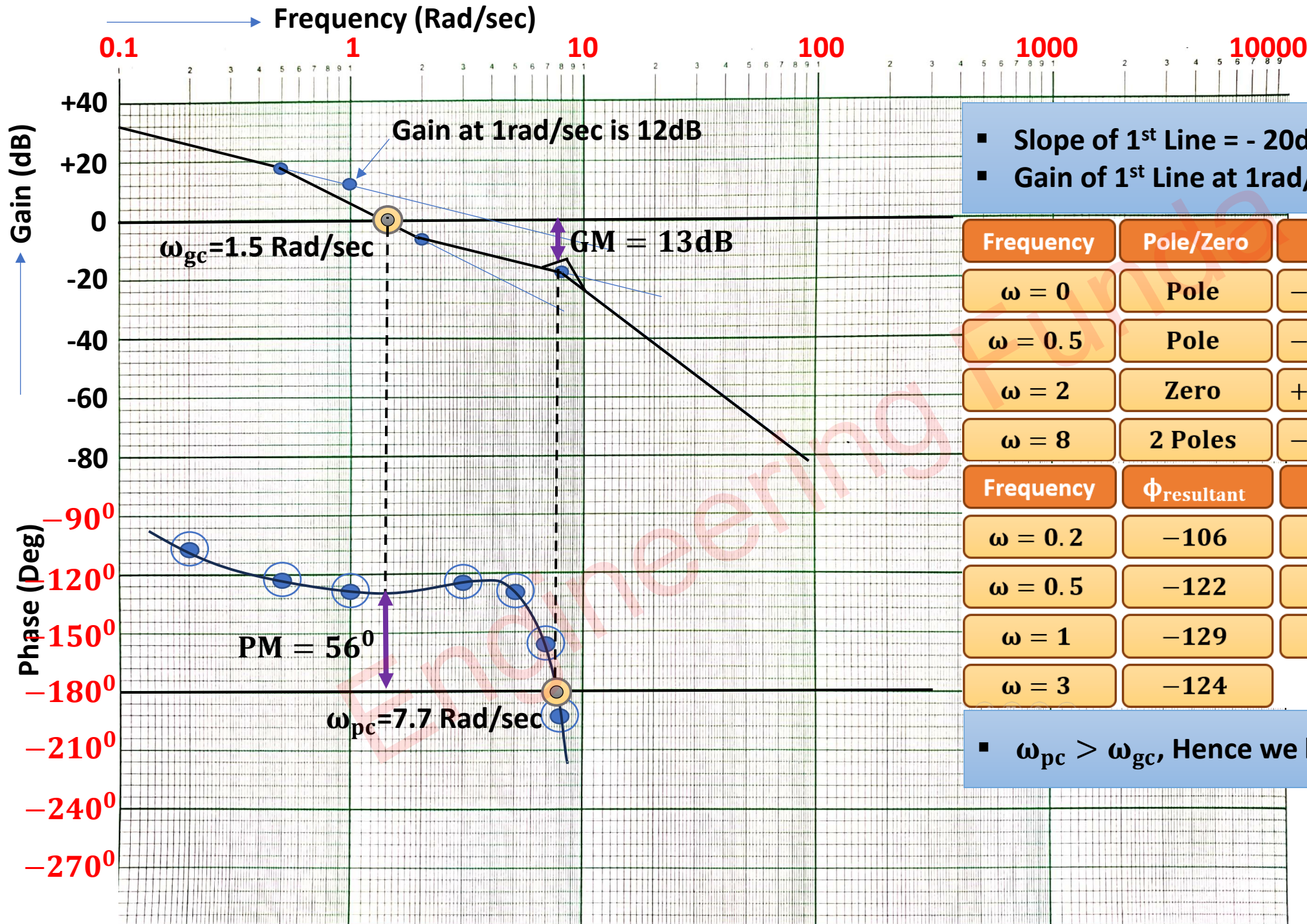
Frequency	Pole/Zero	Slope	Resultant
$\omega = 0$	Pole	-20dB/Dec	-20dB/Dec
$\omega = 0.5$	Pole	-20dB/Dec	-40dB/Dec
$\omega = 2$	Zero	+20dB/Dec	-20dB/Dec
$\omega = 8$	2 Poles	-40dB/Dec	-60dB/Dec

$$\Rightarrow s^2 + 3.2s + 64 = s^2 + 2\xi\omega s + \omega^2$$
$$\Rightarrow \xi = 0.2$$

❑ Step-5: Phase of $\angle G(j\omega)$.

$$\Rightarrow \angle G(j\omega) = \phi_{\text{resultant}} = -90^\circ - \tan^{-1} \frac{\omega}{0.5} - \tan^{-1} \frac{3.2\omega}{64 - \omega^2} + \tan^{-1} \frac{\omega}{2}$$

Frequency	$\phi_{\text{resultant}}$	Frequency	$\phi_{\text{resultant}}$
$\omega = 0.2$	-106	$\omega = 5$	-128
$\omega = 0.5$	-122	$\omega = 7$	-158
$\omega = 1$	-129	$\omega = 8$	-190
$\omega = 3$	-124		



- Slope of 1st Line = - 20dB/Dec.
- Gain of 1st Line at 1rad/sec = $20 \log k = 12\text{dB}$

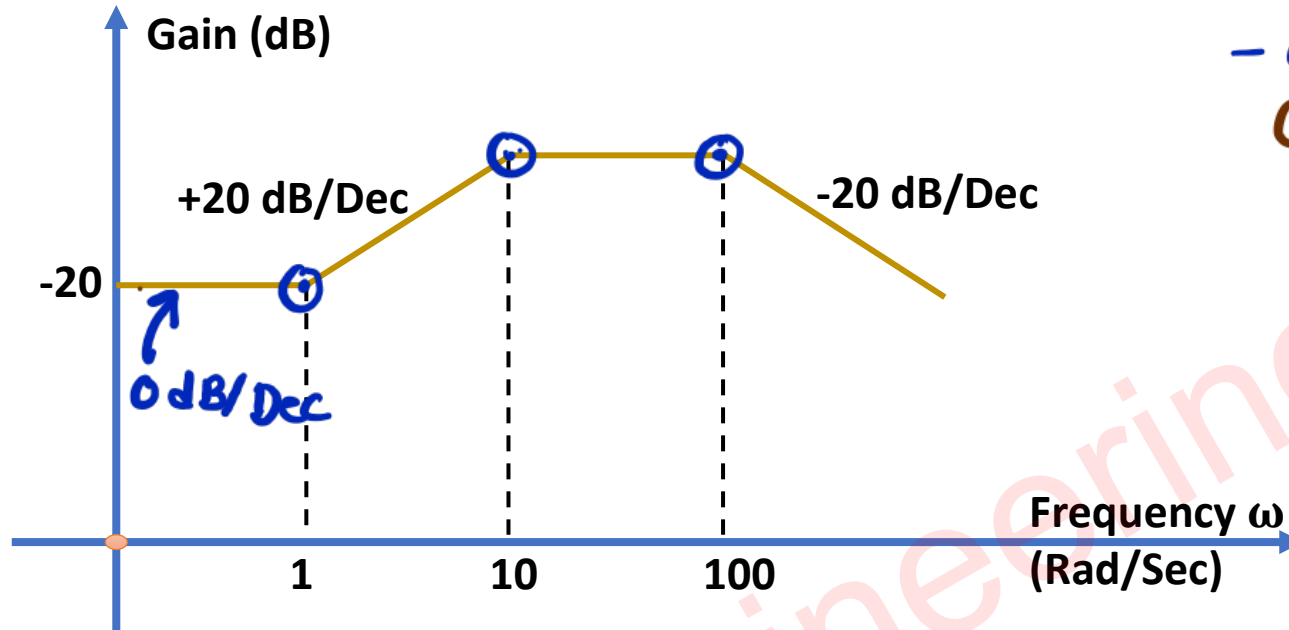
Frequency	Pole/Zero	Slope	Resultant
$\omega = 0$	Pole	-20dB/Dec	-20dB/Dec
$\omega = 0.5$	Pole	-20dB/Dec	-40dB/Dec
$\omega = 2$	Zero	+20dB/Dec	-20dB/Dec
$\omega = 8$	2 Poles	-40dB/Dec	-60dB/Dec

Frequency	$\phi_{\text{resultant}}$	Frequency	$\phi_{\text{resultant}}$
$\omega = 0.2$	-106	$\omega = 5$	-128
$\omega = 0.5$	-122	$\omega = 7$	-158
$\omega = 1$	-129	$\omega = 8$	-190
$\omega = 3$	-124		

- $\omega_{pc} > \omega_{gc}$, Hence we have a Stable System.

Bode Plot Examples

Example 5 — Identify the Transfer Function of the given Gain Plot of the Bode Plot.



$\omega = 1$ (Zero), $\omega = 10$ (Pole), $\omega = 100$ (Pole)

$$\Rightarrow -20 = 20 \log K$$

$$\Rightarrow \log K = -1$$

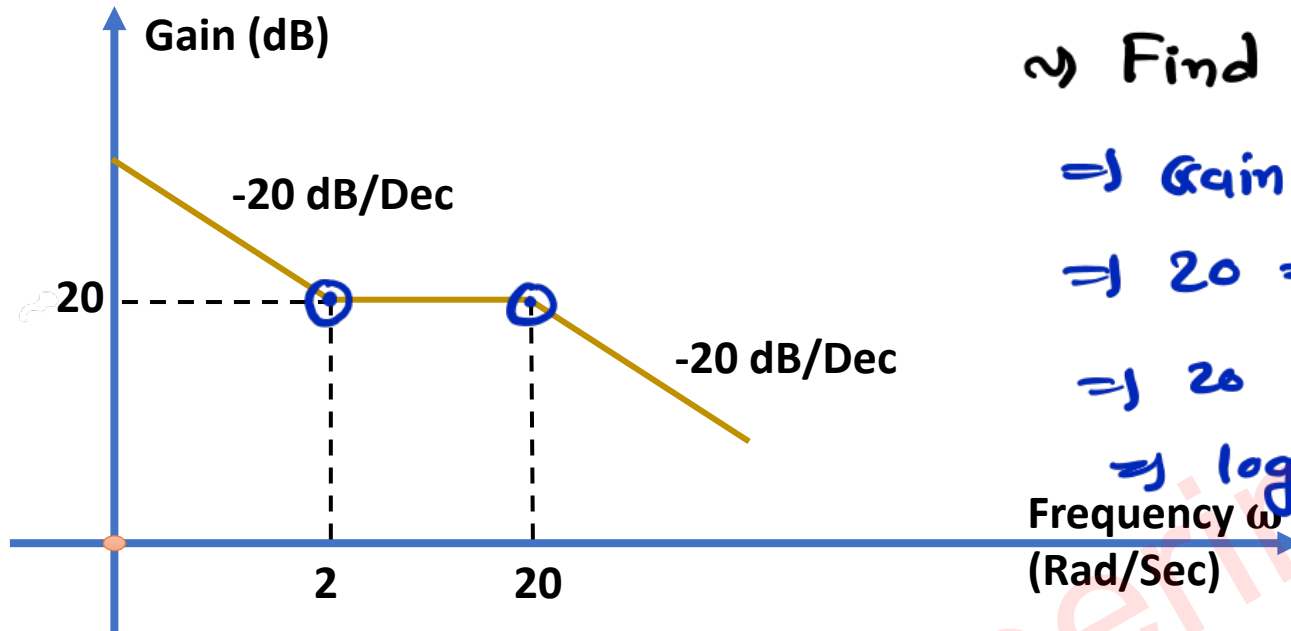
$$\Rightarrow K = 0.1$$

→ Transfer Function

$$\Rightarrow G(s) = \frac{0.1 (1 + s/1)}{(1 + s/10)(1 + s/100)}$$

$$\Rightarrow G(s) = \frac{100 (s + 1)}{(s + 10)(s + 100)}$$

Example 6 — The figure below shows a system's Bode Magnitude plot. The Minimum Phase Transfer Function of the system is given by



⇒ Find Constant K

$$\Rightarrow G_{dB} = 20 \log K + (\text{Slope}) \log \omega$$

$$\Rightarrow 20 = 20 \log K + (-20) \log 2$$

$$\Rightarrow 20 \log K = 26$$

$$\Rightarrow \log K = 1.3 \Rightarrow \boxed{K = 20}$$

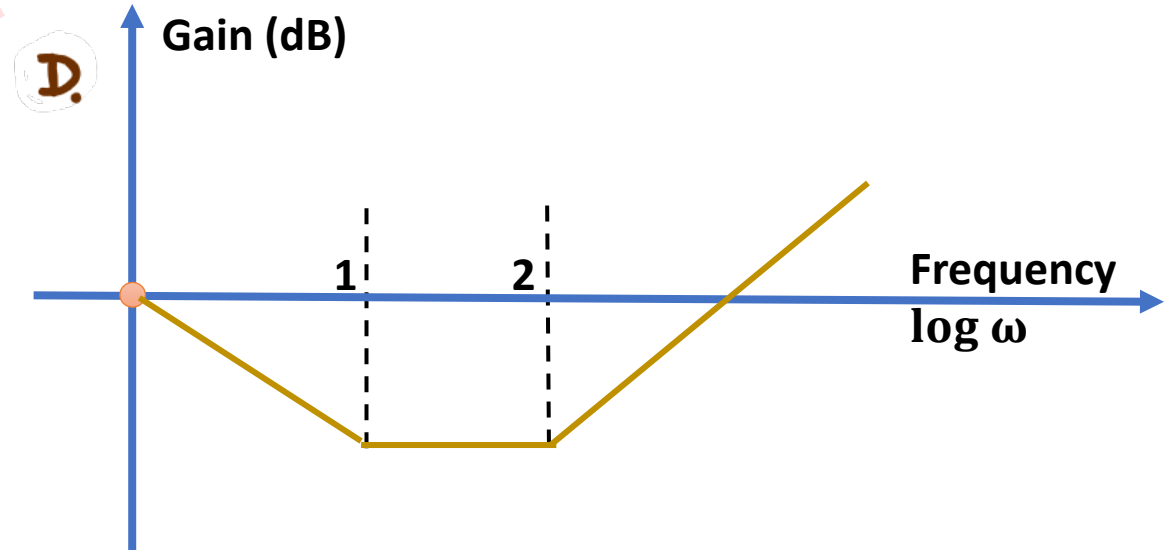
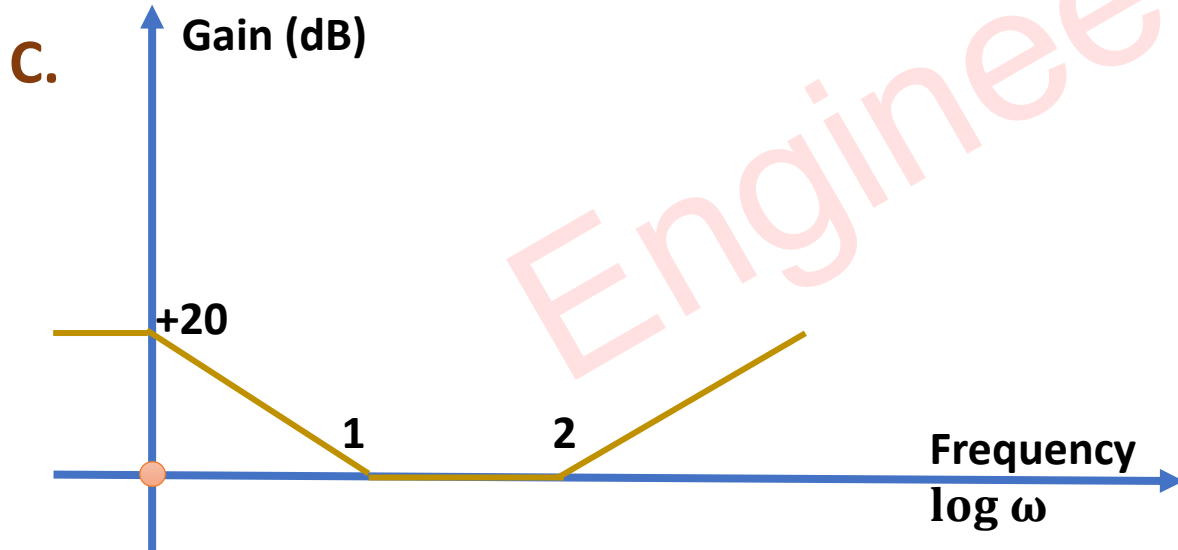
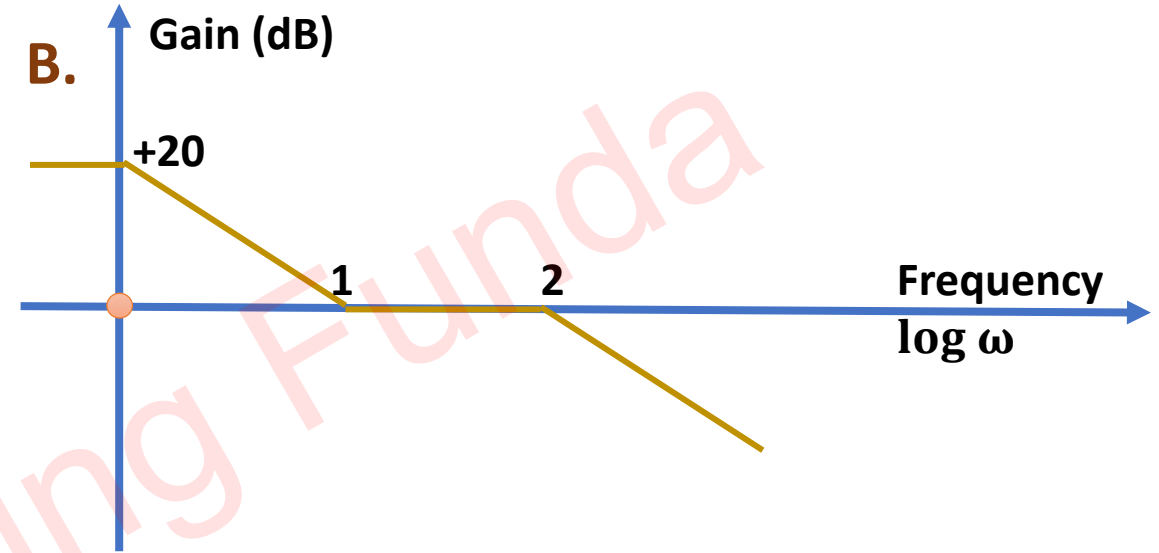
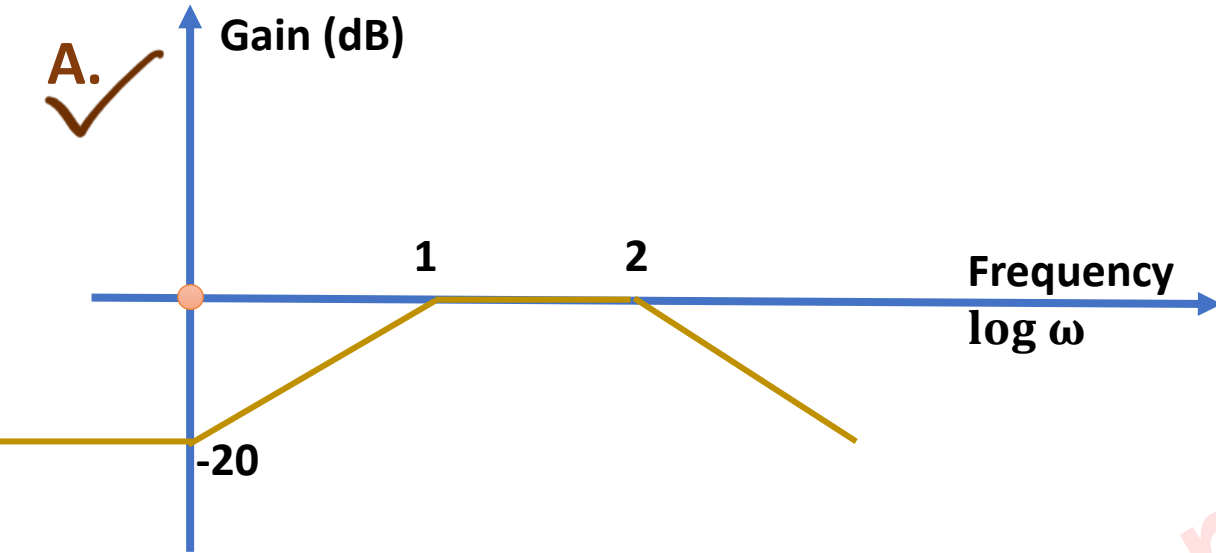
— $\omega = 0$, $\omega = 2$, $\omega = 20$
(Pole) (Zero) (Pole)

⇒ Transfer Function

$$\Rightarrow G(s) = \frac{20 (1 + s/2)}{s (1 + s/20)} = \frac{200 (s + 2)}{s (s + 20)}$$

Example 7 — The Bode Plot of

$$\Rightarrow G(j\omega) = \frac{10^4(1 + j\omega)}{(10 + j\omega)(100 + j\omega)^2}$$



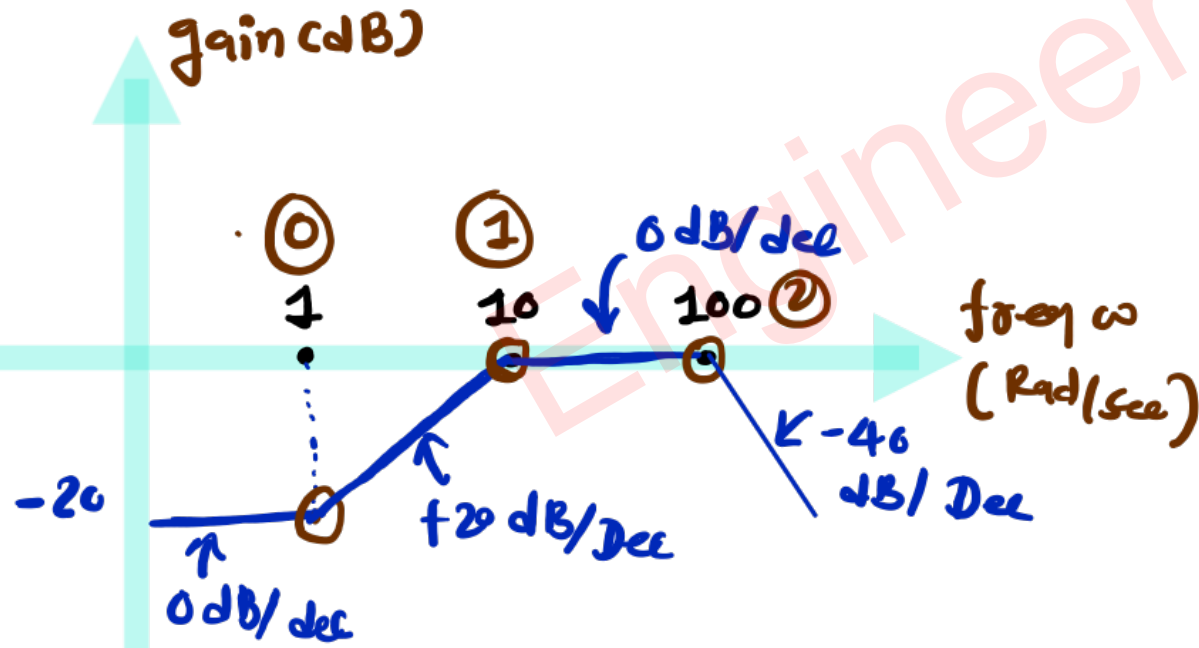
$$\begin{aligned}
 G(j\omega) &= \frac{10^4 (1+j\omega)}{(10+j\omega)(100+j\omega)^2} \\
 &= \frac{10^4 (1+j\omega)}{10 (1+j\omega/10) 10^4 (1+j\omega/100)^2} \\
 &= \frac{0.1 (1+j\omega)}{(1+j\omega/10)(1+j\omega/100)^2}
 \end{aligned}$$

→ gain of 1st line, $\omega = 1 \text{ rad/sec}$

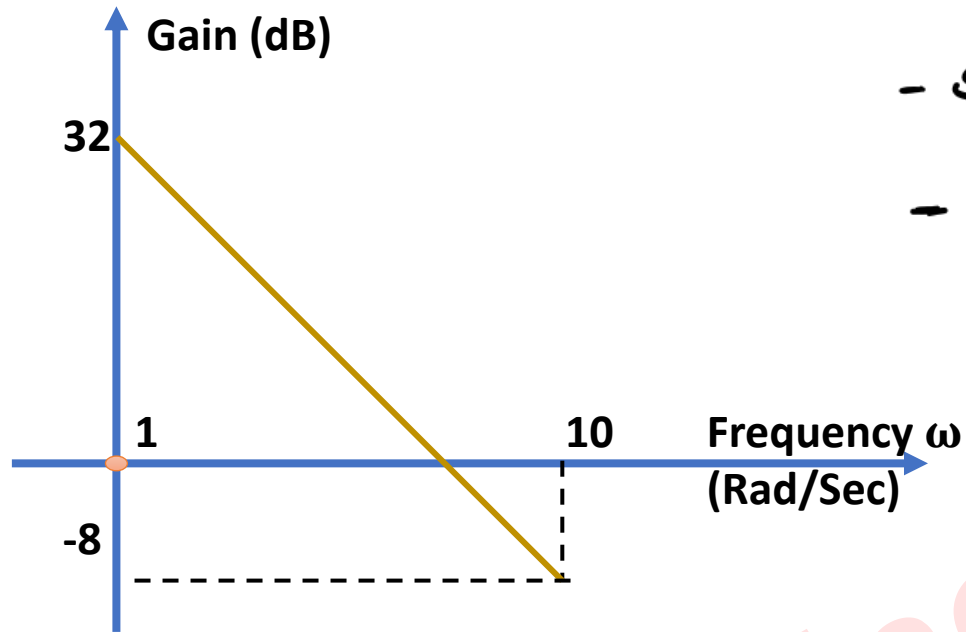
$$= 20 \log K$$

$$= -20 \text{ dB}$$

→ 1st line Slope = 0 dB/Dec



Example 8 — The Bode Plot of a transfer function $G(S)$ is shown in the figure below. The Gain ($20 \log G(S)$) is 32 dB and -8 dB at 1 Rad/sec and 10 Rad/sec, respectively. The Phase is negative for all frequencies. Then $G(S)$ is



- Slope = -40 dB/Dec (2 poles at origin)

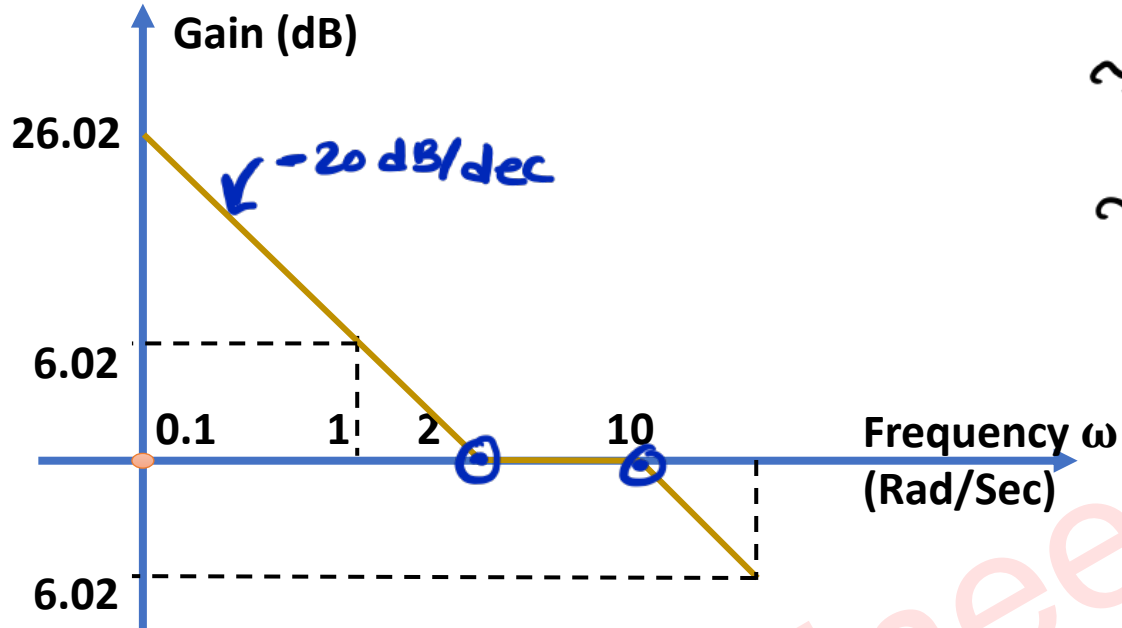
$$- 32 = 20 \log K \Rightarrow \log K = 1.6$$

$$\Rightarrow \boxed{K = 39.81}$$

⇒ Transfer Function

$$\Rightarrow G(S) = \frac{39.81}{S^2}$$

Example 9 — The Bode asymptotic magnitude plot of the minimum phase system is shown in the figure. If the system is connected in a unity negative feedback configuration, the steady state error of the closed-loop system to a unit ramp input is0.5.....



$$\Rightarrow E_{ss} = \frac{1}{K_v} = \frac{1}{2} = 0.5$$

$$\begin{aligned} \Rightarrow K_v &= \lim_{s \rightarrow 0} s G(s) \\ &= \lim_{s \rightarrow 0} \left[\frac{10(s+2)}{s(s+10)} \right] \\ &= 2 \end{aligned}$$

$$\Rightarrow \omega = 0, \omega = 2, \omega = 10$$

(Pole) (Zero) (Pole)

$$\begin{aligned} \Rightarrow 20 \log K &= 6.02 \\ \Rightarrow \boxed{K = 2} \end{aligned}$$

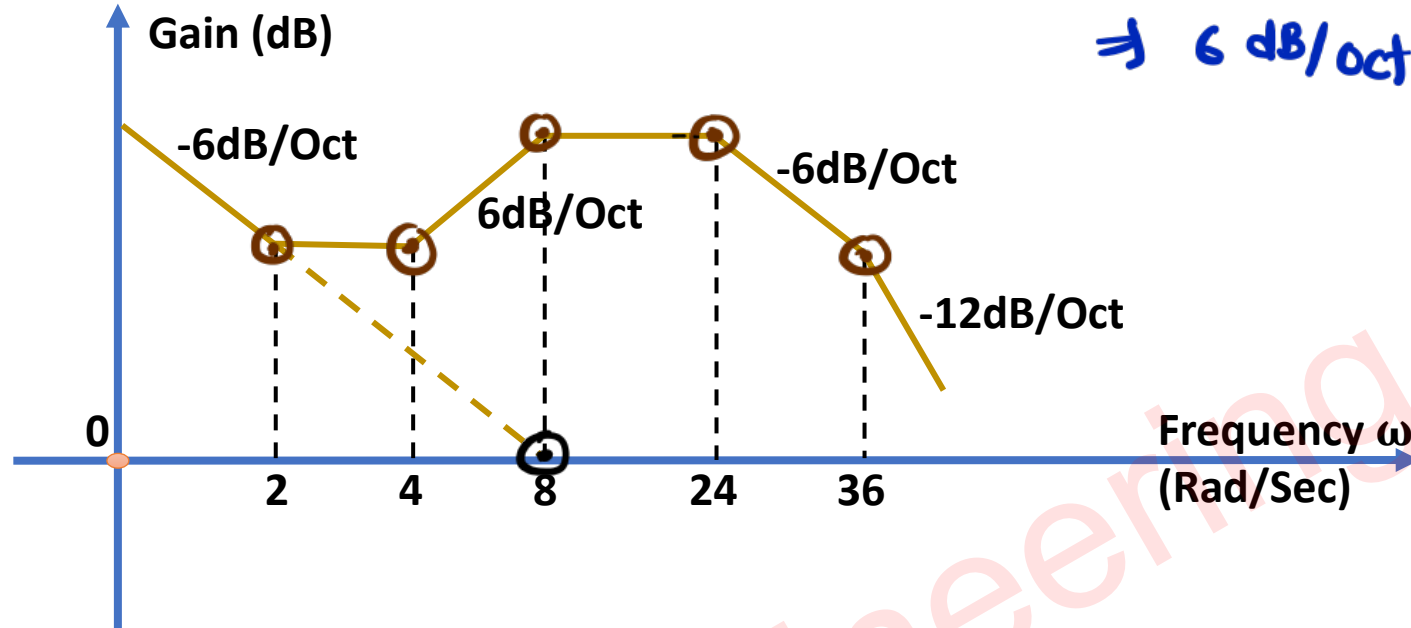
$\Rightarrow$ Transfer Function

$$\Rightarrow G(s) = \frac{2(1 + s/2)}{s(1 + s/10)} = \frac{10(s+2)}{s(s+10)}$$

Example 10 — Find the value of k , a , and b for the bode plot of the transfer function from the plot:

$$\Rightarrow G(S) = \frac{k(1 + 0.5S)(1 + aS)}{S(1 + S/8)(1 + bS)(1 + S/36)}$$

$$\Rightarrow 6 \text{ dB/Oct} = 20 \text{ dB/Dec}$$



$\Rightarrow \omega = 0, \omega = 2, \omega = 4, \omega = 8, \omega = 24, \omega = 36$
 [Pole] [Zero] [Zero] [Pole] [Pole] [Pole]

$\Rightarrow$ Find constant k

$$\Rightarrow \text{Gain} = 20 \log k + (\text{slope}) \log \omega$$

$$\Rightarrow 0 = 20 \log k + (-20) \log 8 \Rightarrow \boxed{k = 8}$$

$\Rightarrow$ Transfer Function

$$\Rightarrow G(S) = \frac{8(1 + S/2)(1 + S/4)}{S(1 + S/8)(1 + S/24)(1 + S/36)}$$

$$\Rightarrow k = 8$$

$$a = 1/4$$

$$b = 1/24$$

Example 11 — The Transfer Function is given by:

$$\Rightarrow G(S) = \frac{5(S + 4)}{S(S + 0.25)(S^2 + 4S + 25)}$$

The Values of the constant gain term and a highest corner frequency of the bode plot are**3.2**..... and**5**....., respectively.

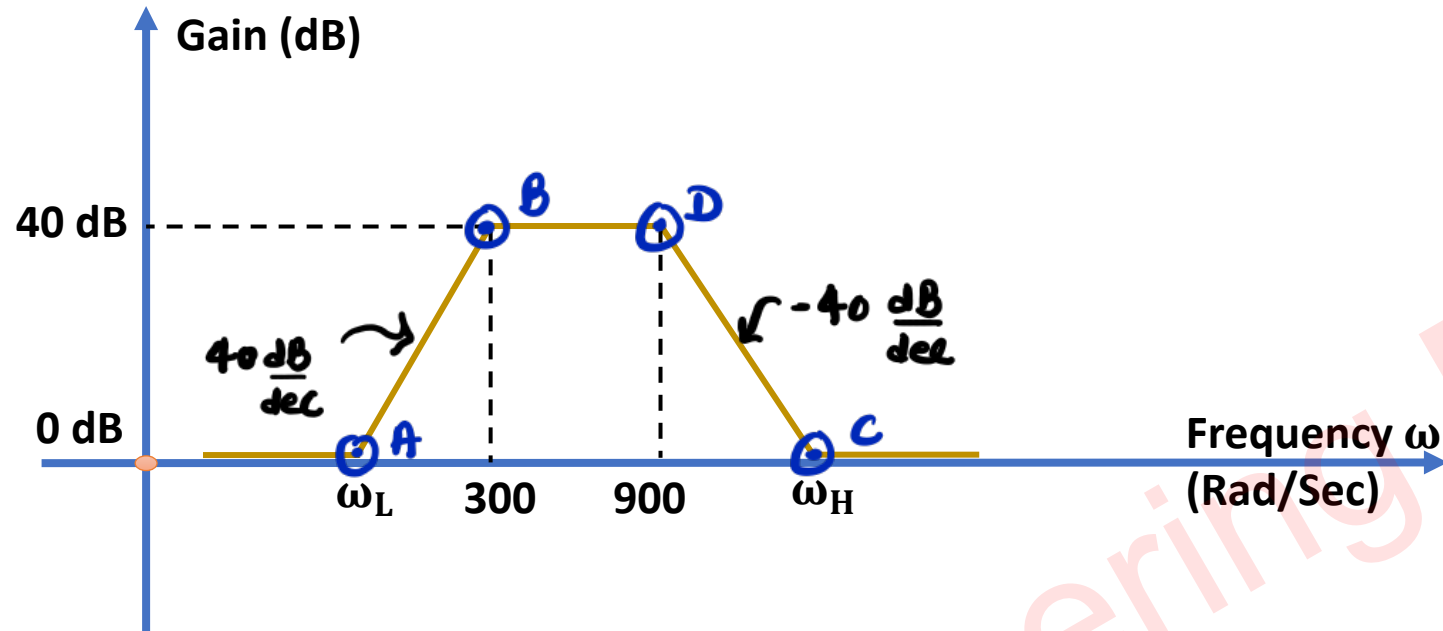
∴ Standard Form of Transfer Function

$$\Rightarrow G(S) = \frac{5(4(1 + S/4))}{S(0.25(1 + S/0.25))(25(S^2/25 + 4S/25 + 1))}$$

$$\Rightarrow G(S) = \frac{3.2(1 + S/4)}{S(1 + S/0.25)(S^2/25 + 4S/25 + 1)} \rightarrow \left(\frac{S^2}{\omega^2} + \frac{2\xi\omega S}{\omega^2} + 1 \right)$$

→ $K = 3.2$, $\omega = 4$ (Zero) , $\omega = 0$ (Pole) , $\omega = 0.25$ (Pole) , $\omega = 5$ (Poles 2)
Highest

Example 12 — Consider the plot shown in the figure. Assume all the poles and zeros are real-valued. Find $\omega_H - \omega_L$.



$$\sim \text{Slope } AB = \frac{G_B - G_A}{\log \omega_B - \log \omega_A}$$

$$\Rightarrow \cancel{40} = \frac{\cancel{40}}{\log \left(\frac{300}{\omega_L} \right)}$$

$$\Rightarrow \log \left(\frac{300}{\omega_L} \right) = 1$$

$$\sim \text{Slope } CD = \frac{G_D - G_C}{\log \omega_D - \log \omega_C}$$

$$\Rightarrow \cancel{-40} = \frac{\cancel{40}}{\log \left(\frac{900}{\omega_H} \right)}$$

$$\Rightarrow \log \left(\frac{900}{\omega_H} \right) = -1$$

$$\Rightarrow \frac{300}{\omega_L} = 10$$

$$\boxed{\Rightarrow \omega_L = 30 \text{ rad/sec}}$$

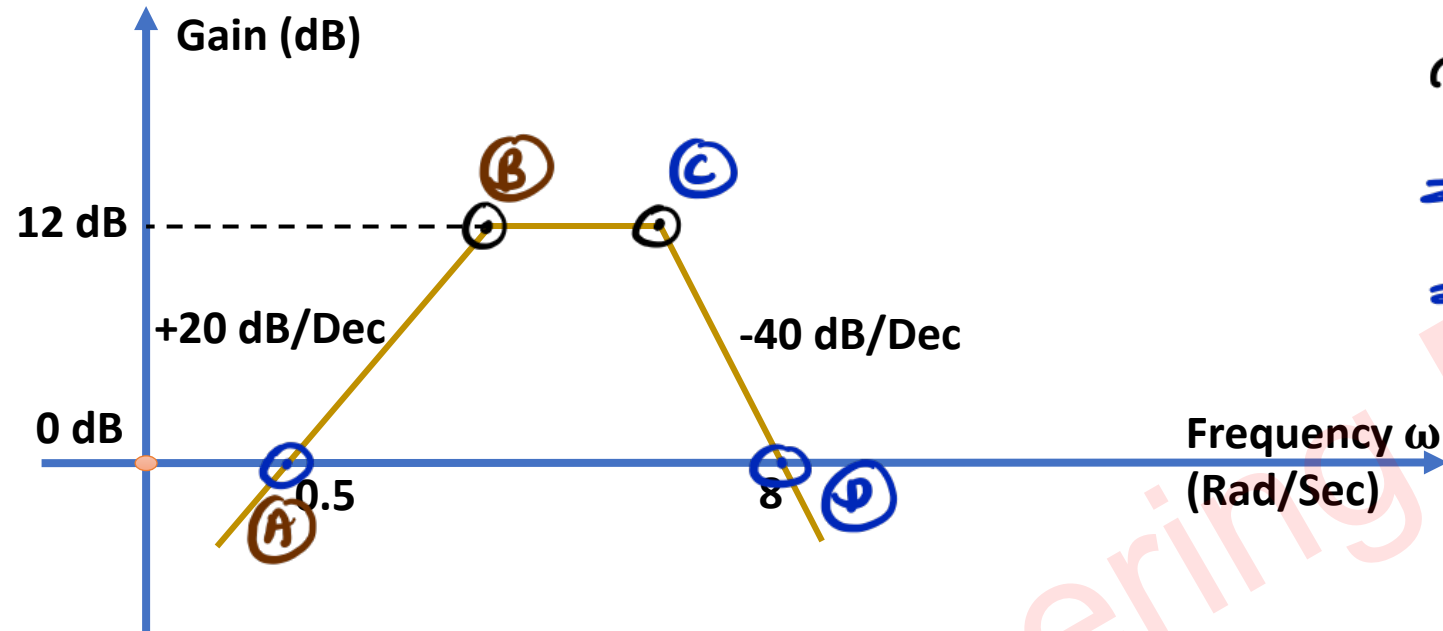
$$\Rightarrow \frac{900}{\omega_H} = 0.1$$

$$\boxed{\Rightarrow \omega_H = 9000 \text{ rad/sec}}$$

$$\Rightarrow \omega_H - \omega_L = 9000 - 30 = 8970 \text{ rad/sec}$$

Engineering Funda

Example 13 — Consider the plot shown in the figure. Assume all the poles and zeros are real-valued. Find the Transfer Function.



~ $\omega = 0$ [Zero]

$$\sim \text{Slope AB} = \frac{G_A - G_B}{\log \omega_A - \log \omega_B}$$

$$\Rightarrow 20 = \frac{0 - 12}{\log \left(\frac{0.5}{\omega_B} \right)}$$

~ Find Constant K

$$\Rightarrow G_{\text{min}} = 20 \log K + (\text{Slope}) \log \omega$$

$$\Rightarrow 0 = 20 \log K + (20) \log 0.5$$

$$\Rightarrow \boxed{K = 2}$$

$$\sim \text{Slope CD} = \frac{G_C - G_D}{\log \omega_C - \log \omega_D}$$

$$\Rightarrow -40 = \frac{12 - 0}{\log \left(\frac{\omega_C}{8} \right)}$$

$$\Rightarrow \log \left(\frac{\omega_C}{8} \right) = -\frac{12}{40}$$

$$\Rightarrow \log \left(\frac{0.5}{\omega_B} \right) = -0.6$$

$$\Rightarrow \boxed{\omega_B = 2 \text{ rad/sec}}$$

[Pole]

$$\boxed{\Rightarrow \omega_C = 4 \text{ rad/sec}}$$

[2 poles]

~ Transfer Function

$$\Rightarrow G(s) = \frac{2s}{(1+s/2)(1+s/4)^2} = \frac{64s}{(s+2)(s+4)^2}$$